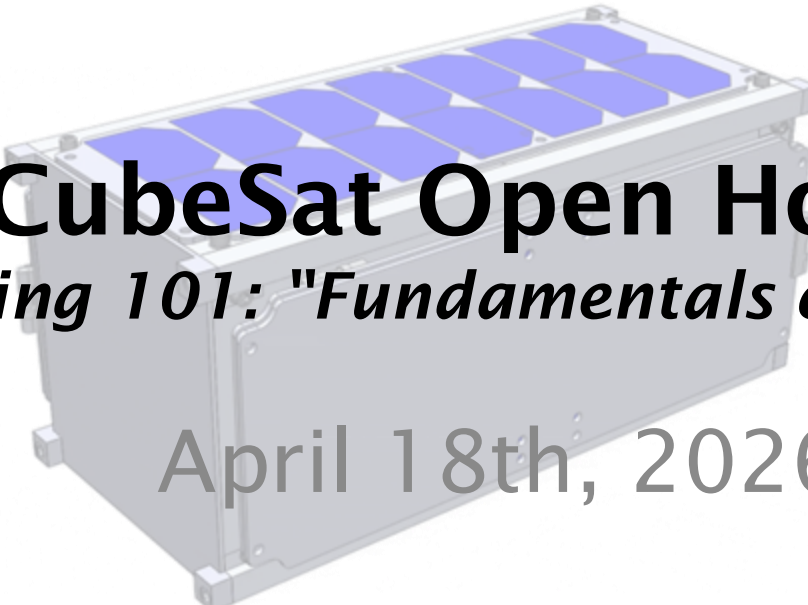


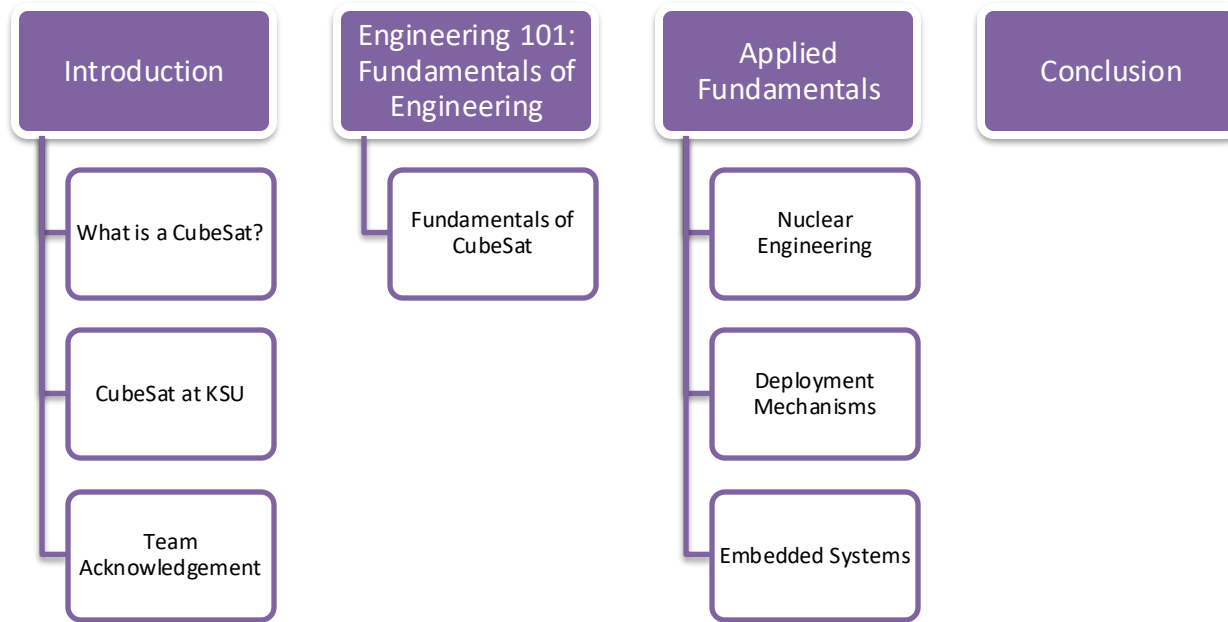


CubeSat Open House

Engineering 101: "Fundamentals of Engineering"

April 18th, 2026

Agenda



What is a CubeSat?

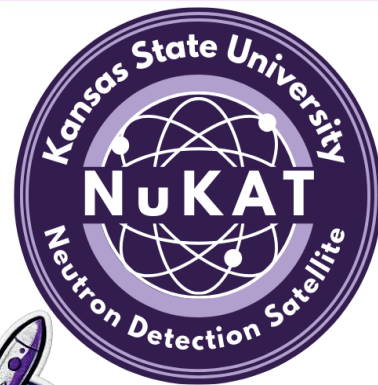
- Satellites built with a standardized, scalable form factor
 - 10 x 10 x 10-11.35 cm "cube"
 - Popular sizes: 1U, 2U, 3U, 6U, & 12U
- Low-cost path to satellite industry
- Used in university, government, and commercial settings
 - Science, tech demos, & earth observation



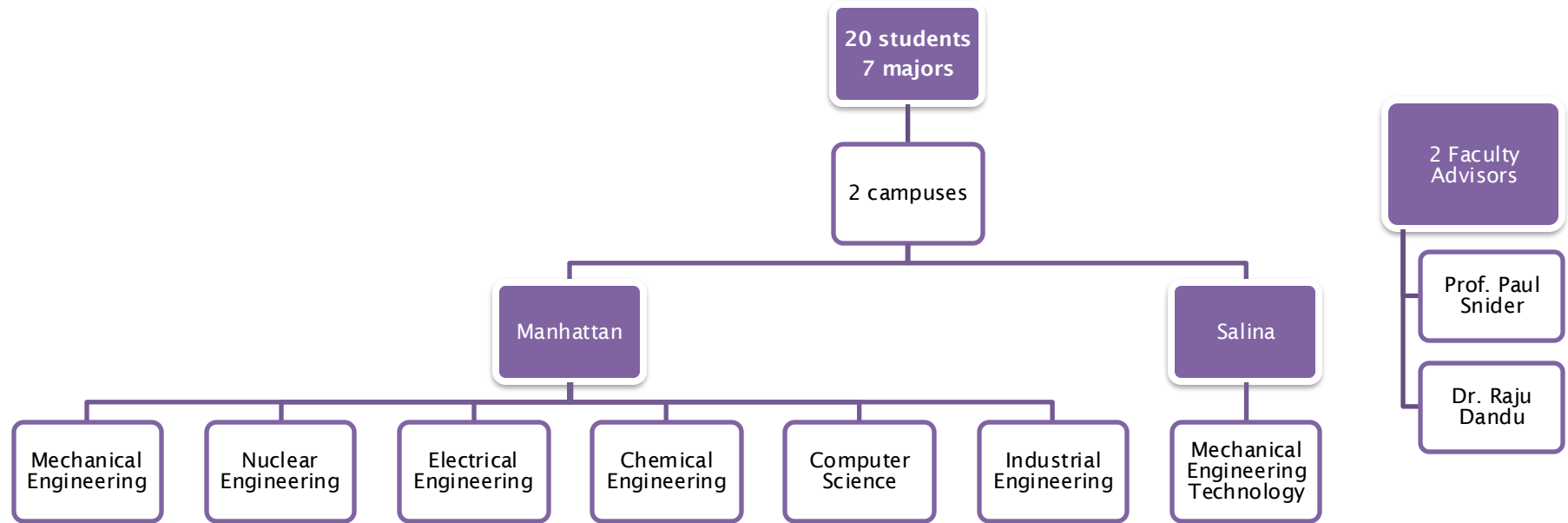
Figure: CubeSat (Cal-Poly – CubeSat 101)

CubeSat at Kansas State University

- Idea by Bailey Walke & Paul Snider
 - Project Incubate in Fall 2024
 - Class started Spring 2025
- Backed by Wildcat Space Program
 - Founded Spring 2026
- Led by students, guided by faculty
- Goal: launch 2 CubeSats by 2030
 - NuKat (primary) PHAT-Sat (secondary)

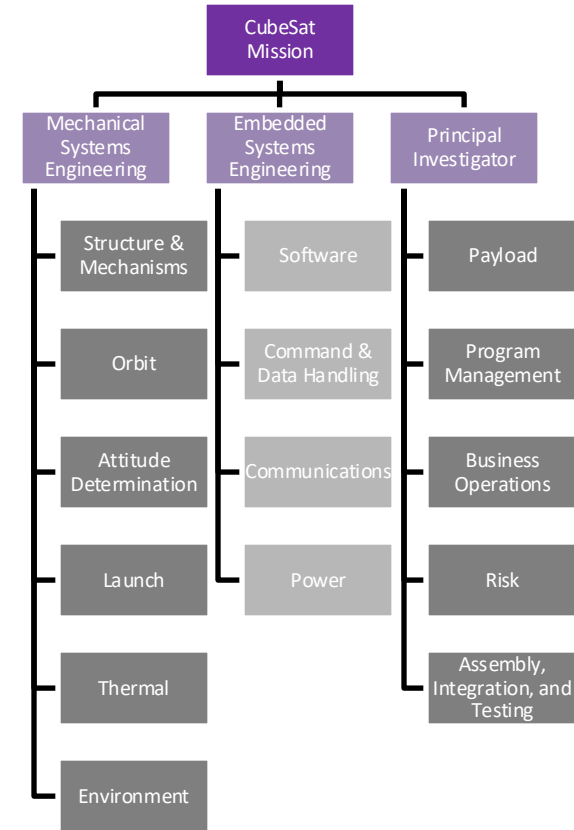


Team Acknowledgement



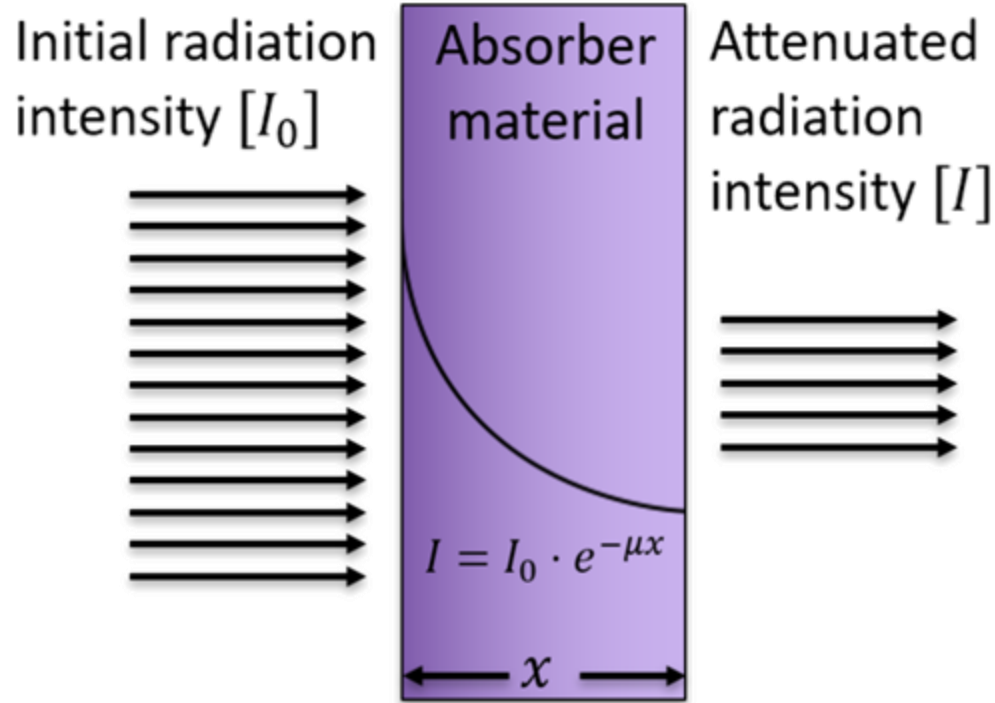
Engineering 101: Fundamentals of CubeSat

- Interdisciplinary engineering
- Engineering management
 - Systems engineering
 - Risk management
 - Finances and timeline
- Idea generation
- Project reviews
- Industry networking and consultation



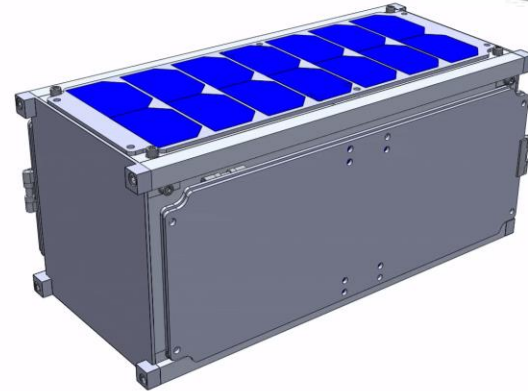
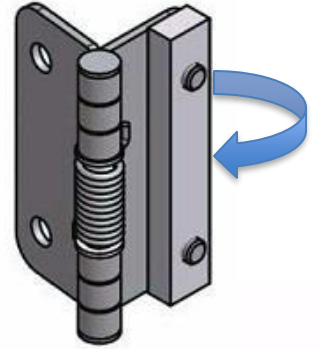
Applied Fundamentals - Nuclear Engineering

- Radiation Shielding
 - Beer-Lambert Law: $I = I_0 \cdot e^{-\mu x}$
- Space radiation
 - Directly ionizing radiation ($p^\pm, \beta^\mp, \gamma, \alpha, GCRs$)
 - Indirectly ionizing radiation ($n^0, X\text{ rays}, \gamma\text{ rays}$)
- Protecting electronics



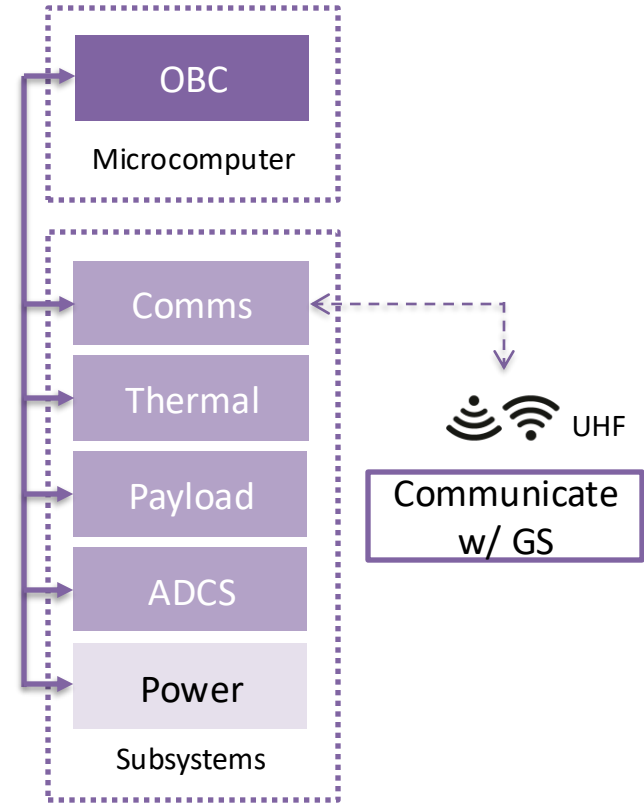
Applied Fundamentals – Deployment Mechanisms

- Spring-loaded hinges to deploy solar panels
- Must provide enough torque to overcome **friction of hinges** and **tension from wiring**, *without* slamming components
- Hooke's Law: $T = k\theta$
- Sizing spring constant k :
 - Too small $\rightarrow$ panels don't deploy
 - Too large $\rightarrow$ panels deploy too aggressively

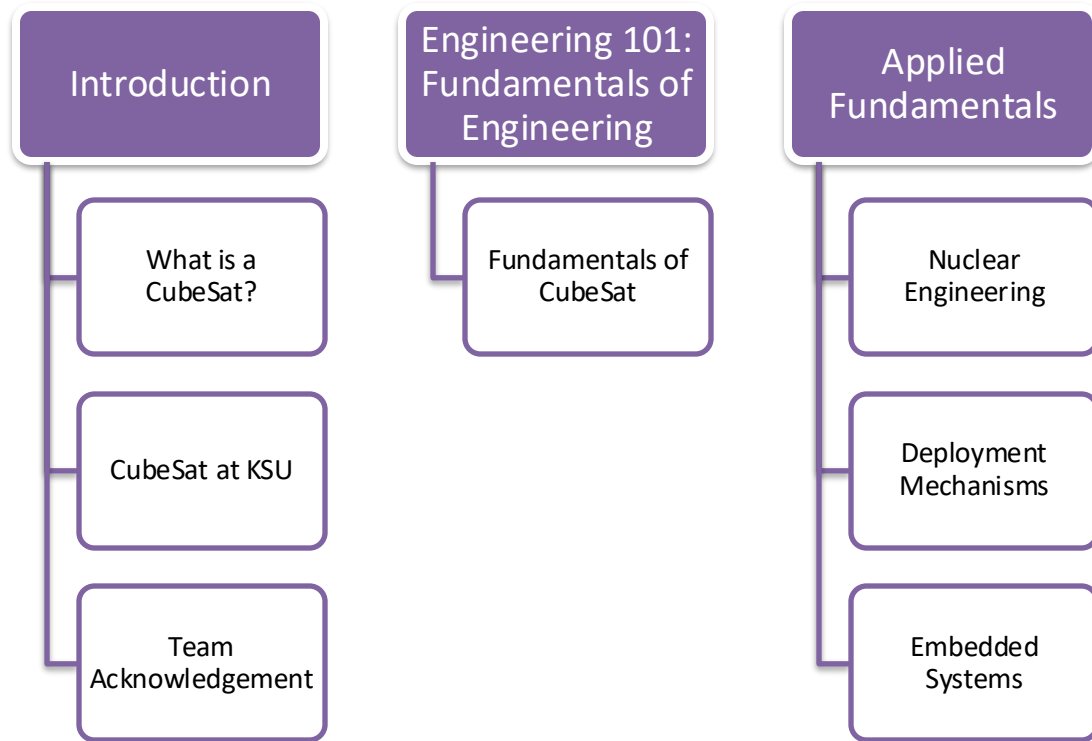


Applied Fundamentals – Embedded Systems

- Hardware Evaluation
 - Component selection trade studies
- Hardware Interface Diagrams
 - Wiring diagrams
 - Wireless/serial communications
 - Packet/frame design
- Software Architecture
 - FPGA/RISC-V SoC co-design
- Software Library Identification
 - NASA Core Flight System (cFS)



Conclusion





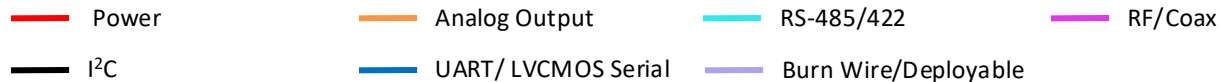
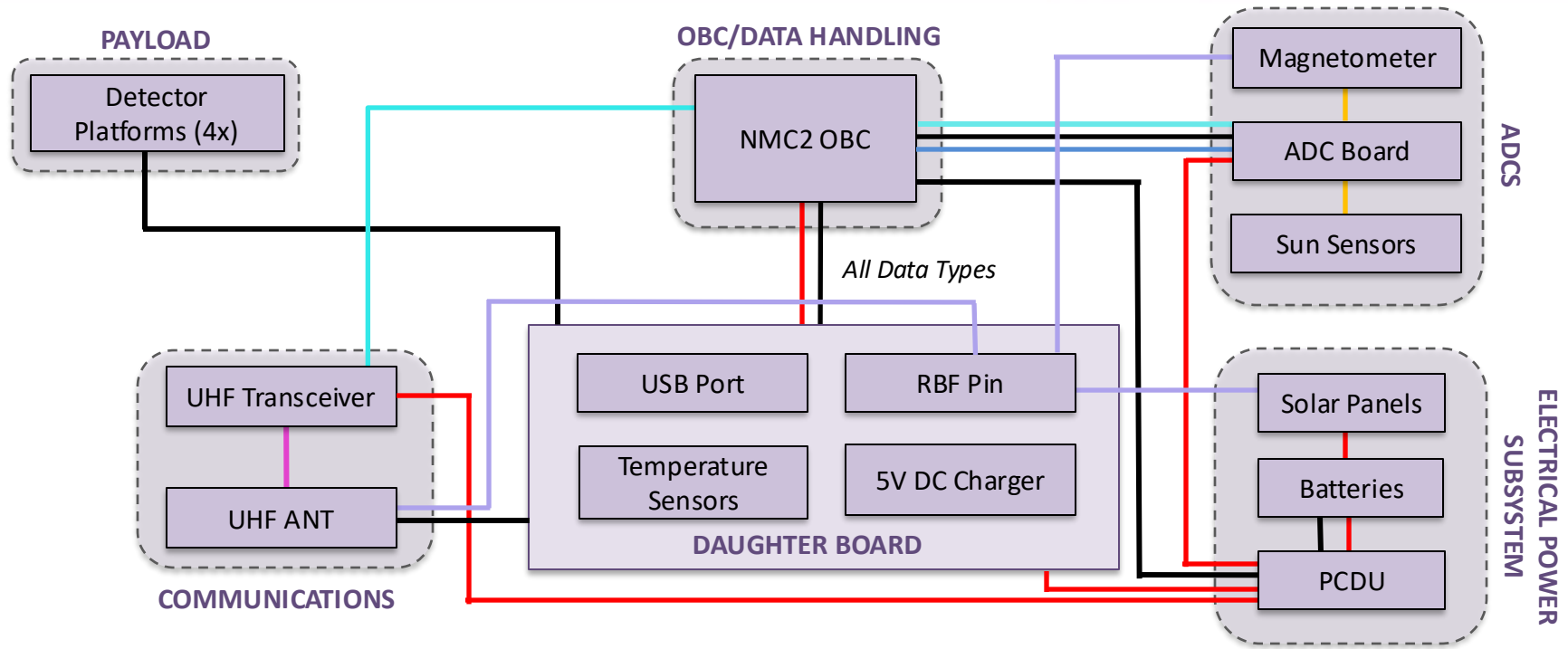
Questions?



Backup Slides

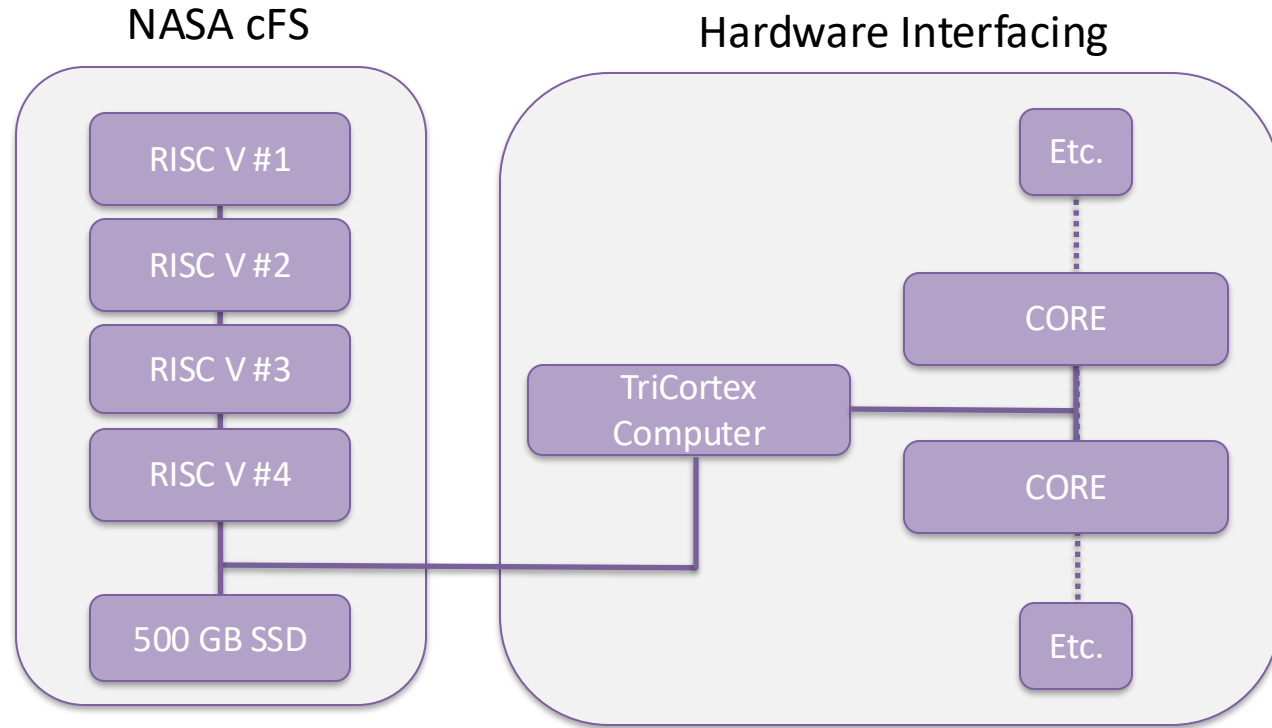
Information we would love to present, but don't have time for... 😊
Use as needed!

Hardware Interface Diagram



Software Architecture Diagram

- NextAge NMC2
 - Polarfire SoC
 - FPGA Fabric
 - Programmable with TriCortex
 - Can be split into many individual cores
 - 4 Risc V Cores
 - Handles individual tasks in parallel
 - Runs NASA cFS
 - FPGA HAL
 - RISC V Main Compute



Chemical Engineering Fundamentals

- Material properties
- Material behavior in space
- Shielding

Electrical Engineering Fundamentals

- $P = V * I$
- $V = I * R$
- Power Budget
- Link Budget
- Communications -> Serial, RF, Analog, Digital, etc.

Computer Science Fundamentals

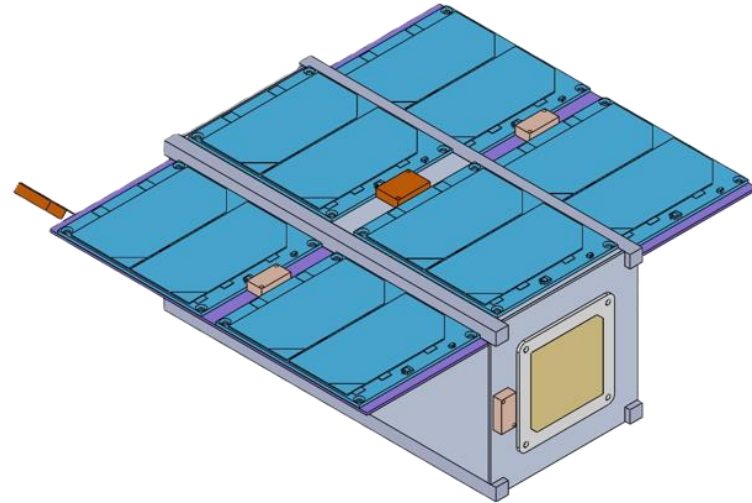
- Software Development Plan
- Software Architecture
- Software Testing Plan
- Library Identification
- Hardware Evaluation

Industrial Engineering Fundamentals

- Risk assessment
- Budget
- Scheduling
- Ground Station
 - K-State Radio Station

Mechanical Engineering Fundamentals

- CAD design
- Deployable mechanisms
- Design for manufacturing
- Structural and thermal analysis



Mechanical Engineering Technology Fundamentals

- Risk assessment
- Scheduling
- Dimension/Tolerance Verification
- Hands-on Testing
- Manufacturing Processes

Nuclear Engineering Fundamentals

- Computer simulations
- Neutron energy moderation
- Neutron detector modeling
- Radiation Shielding

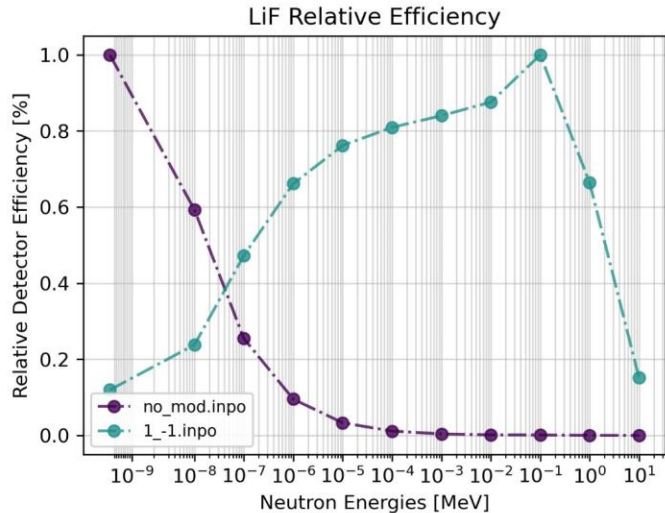


Figure: Detector relative efficiencies for **moderated** and **unmoderated** detector platforms

